ILLUSTRATION: Mahesh Benkar

Denying you Service

Plant a tiny ticking time bomb that has been fine tuned to attack, on the Internet. Watch it lie in wait for a single instruction. Traffic on the Internet comes to a crashing halt when the moment comes. The show has just begun—or should we say, ended...

Picture this: you're heading off to the beach on a fine sunny day, and you've just pulled down the top of your convertible. As you leisurely cruise down the highway, you can't help but feel good when you think about the rest of the day. That is, until you face the mother of all traffic jams right up ahead, and there soon seems no way for you to progress. Now apply that analogy to a 'Denial of Service' (DoS)' attack on the Internet: the information autobahn—that lifeline of all services online—that you are cruising on, can turn out to be worse than that demolition derby you saw on TV.

In the networked world, a single click of a button can turn off the gas. That's as simple as it's getting to be, to attack the Internet.

Denying you access

A Denial of Service attack is one in which a remote machine or network is targeted such that its ability to conduct normal and legitimate business is tied down by swamping it with bogus requests for information, and utilising all the available machine and network resources. Distributed Denial of Service (DDoS) of service attacks initiate hostile action from several machines having diverse, and possibly rapidly changing, origins.

A DDoS attack could involve large numbers of slave or compromised computers, from between a few hundred to several thousand. The attacks are usually well planned and smoothly orchestrated, take place in discrete steps and generally take advantage of some known vulnerability in the targeted machines. The first phase involves rapidly scanning a very large number of diversely located hosts to find out the machines that respond to an existing vulnerability. This could be some previously known vulnerability, or some new, unknown one—in the software or hardware setup of the machines being scanned.

Once all the hosts have been probed, the machines that have responded favourably—that is, those that still have